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Roll No

FT-3004-CBGS

B.E. III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

**Heavy Vehicles and Automobile Engineering and
Safety**

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe with suitable sketches the combustion phenomenon in C.I. engines and explain the two phases of combustion.
b) What are the main parts of a Multi-plate clutch? Describe the function of each part.
2. a) What are the functions of suspension in a motor vehicle?
b) What are the requirements of suspension system?
3. a) Explain pollution control techniques for automobile. Explain any two.
b) What is Super charging? Describe different types of fuel injectors.
4. a) Explain the four stages of combustion in a diesel engine.
b) What is the need of gear box? Explain different types of gear box.

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PTO

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5.
 - a) What is magneto and battery ignition?
 - b) How engines are tested on dynamometer?
6.
 - a) What are different types of oil pumps?
 - b) How working of power steering is different from manual steering?
7. Explain the different types of lubricating system.

OR

- a) Explain how the king-pin inclination produces directions stability.
 - b) Describe with a neat sketch, the hydraulic break system of a car?
8. Write short note on :
 - a) CMV rules
 - b) Breaching, PTO
 - c) Supercharging
 - d) Catalytic converter.

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